Team Name: 5G-Team
Student Name: Aoyu Gong, Arman Maghsoudnia, Eduard Vlad, Raphael Cannatà
SCIPER: 306476, 370707,374438, 358968

Data Visualization – COM-480

Milestone II

---

1 General View of the Website

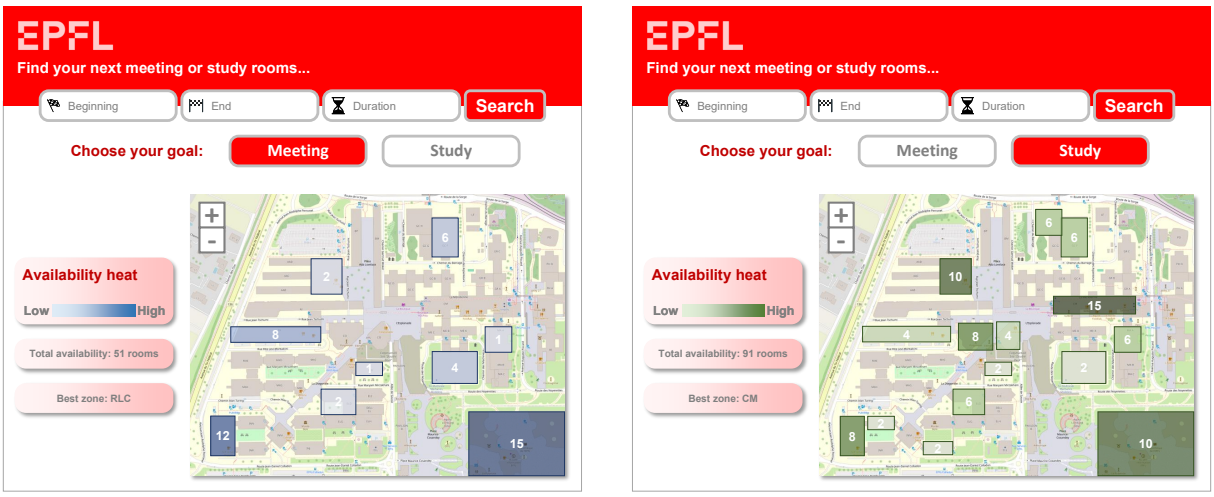
Our project consists of an interactive web application designed to help users find available rooms at EPFL for studying or meetings. The website integrates real-time occupancy data with an intuitive visualization that combines spatial and temporal exploration. Users begin by specifying a **time interval**, including (1) a *starting time*, (2) an *ending time*, and (3) a *duration*, and by selecting a **goal**, either (1) *meeting* or (2) *studying*. Based on these inputs, the website retrieves and displays room availability across the campus.

The website is structured around two hierarchical views. The main page is a map-based visualization that provides a global overview of room availability across EPFL. From this overview, users can interact with buildings directly and access a second, more detailed view. This second view is a list-based exploration that presents individual rooms inside a building and their availability over time. This design follows a natural exploration workflow: users first identify the desired building on the map, and then compare rooms in detail.

2 Main Page: Map-Based Visualization

As illustrated in Figure 1, the main page of the website is an interactive map of the EPFL campus that visualizes room availability at the building level. At the top of the page, users can define their query by selecting a beginning time, an end time, and a duration, as well as choosing between meeting and study modes. These controls directly influence the data displayed on the map.

Each building on the map is represented as a block containing a numerical indicator that shows the number of available rooms within that building for the selected time interval. In addition to the numerical indicator, availability is also represented using color. In meeting mode (see Figure 1a), a blue color scale is used, while in study mode (see Figure 1b), a green color scale is applied. In both cases, color intensity reflects the level of availability, with darker shades indicating a higher number of available rooms. A legend is provided to help users interpret this indicator, ranging from low to high availability.



(a) Meeting mode.

(b) Study mode.

Figure 1: Main page sketch: map-based visualization.

Alongside the map, summary indicators provide additional context, including the total number of available rooms across the campus and the building zone with the highest availability. These elements allow users to quickly understand the overall situation without needing to inspect each building individually. The map also supports interaction, such as zooming and clicking. When a user clicks on a building, they transition to a more detailed view that focuses on that specific location. This interaction establishes the map as the main exploratory interface, guiding users toward areas of interest.

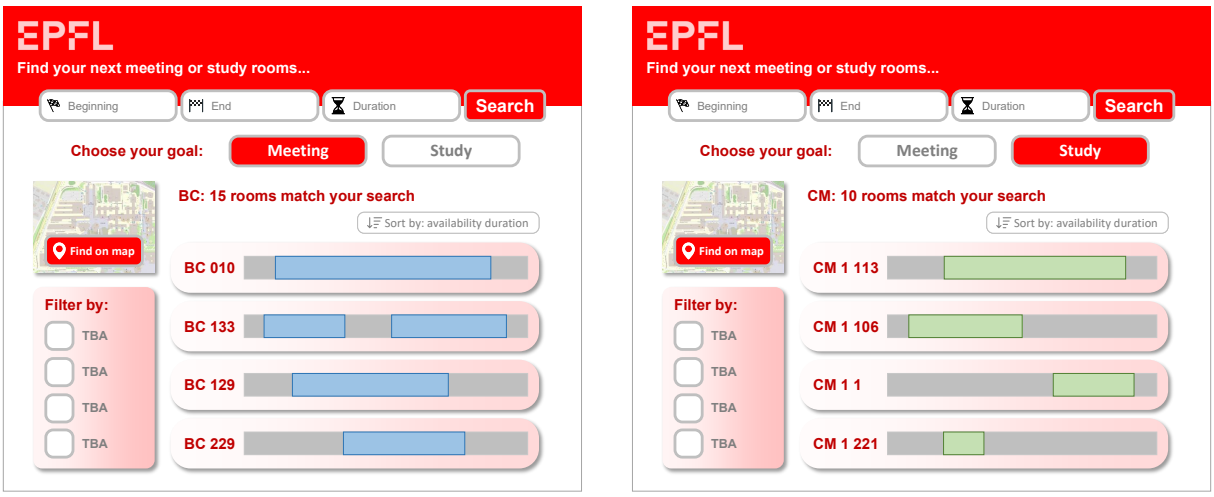
3 Sub Page: List-Based Exploration

As illustrated in Figure 2, the second page provides a detailed, room-level view of availability within a selected building. This page is accessed by clicking on a building on the main page. While the map offers a global overview, this interface focuses on a precise comparison between individual rooms.

At the top of the page, the same controls for time selection and goal selection are available, ensuring consistency between views and allowing users to refine their query if needed. On the left side, a filtering panel offers additional options, such as switching back to the map view or applying further constraints. These filters are designed to be extended in the future with attributes such as room capacity or equipment.

The main component of this page is a list of rooms that match the user's criteria within the selected building. Each room is displayed with a horizontal timeline representing its availability over the selected time interval. Available periods are shown in blue, while occupied periods are shown in gray. The length and position of these segments encode when and for how long the room is available. This allows users to quickly identify rooms that match their specific needs, such as continuous availability for a given duration.

The list can be sorted based on availability duration, ensuring that the most relevant



(a) Meeting mode. (b) Study mode.

Figure 2: Sub page sketch: list-based exploration.

rooms appear first. This detailed view complements the map by providing the temporal resolution necessary for room search. While the map helps users find where to look, this page helps them decide which room to choose.

4 Tools and Methods

The project is implemented as a client-side web application built with `HTML`, `CSS`, and `JavaScript`, where all interaction logic executes in the browser. Users specify a search time window (start and end time), a duration, and room-type filters. These parameters are parsed and validated locally and then used to recompute availability metrics. Spatial visualization is provided through `Leaflet` on top of `OpenStreetMap` tiles. EPFL buildings are rendered as interactive polygon layers, whose geometries are resolved from `OpenStreetMap` references and cached in the browser to reduce repeated requests. Each building polygon is styled as a heatmap. The color represents the proportion of rooms that can host a continuous free interval of at least the requested duration within the selected time window.

The data pipeline combines local metadata with live occupancy data. Local metadata provides building identifiers and room/type information. Live occupancy data is fetched at runtime from a remote HTTP endpoint in JSON format. After loading, occupancy events are indexed by normalized room identifiers and transformed into time intervals suitable for computing (i) building-level availability scores and (ii) per-room timelines. The building side panel is generated dynamically in `JavaScript`. It lists rooms for the selected building and separates them into available and unavailable groups for the chosen time window. For each room, a 24-hour timeline is displayed. This timeline uses positioned segments to show occupied periods and the free intervals between them.

5 Basic Skeleton

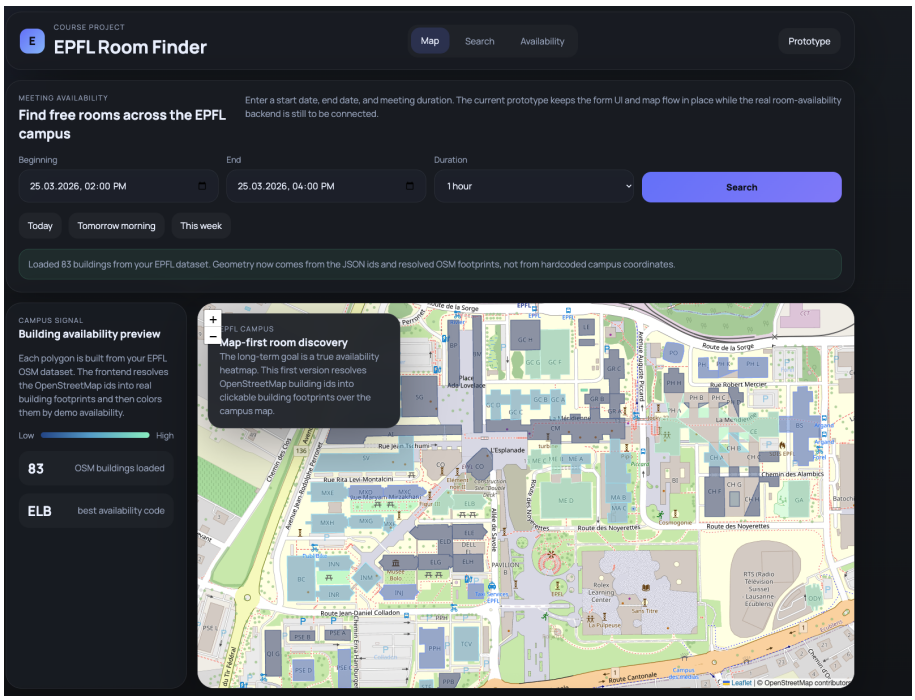


Figure 3: Basic skeleton.

As shown in Figure 3, the current prototype represents the basic skeleton of our website, focusing on establishing the main interaction flow between user input and spatial visualization. As shown in the interface, users can define a query by selecting a starting time, ending time, and desired duration, which serves as the primary filtering mechanism for room availability. The central component of the page is a map-based visualization of the EPFL campus, where buildings are rendered using real geographic data derived from OpenStreetMap. Each building is represented as an interactive polygon, laying the basis for future integration of availability data. Although the current version uses a placeholder, it already demonstrates the intended workflow: users input their requirements, observe availability on the map, and interact with buildings to explore further details. This prototype validates the core design choice of our website.

6 Extra Ideas

There are several ways the visualization could be improved beyond its current version.

One possible improvement is to refine the visual encoding of availability on the map. Currently, buildings are represented using a single color scale based on the number of available rooms. This could be improved by using more informative encodings, such as normalizing availability by building capacity or using bivariate color schemes to simultaneously represent multiple variables (e.g., number of free rooms and average availability duration). This way, users could more easily compare buildings of different sizes.

Another area to improve is the temporal visualization. While the current design relies on fixed time intervals and per-room timelines, the interface could support continuous exploration over time. For example, an interactive time slider or animated transitions could allow users to observe how availability evolves over time. This would make it easier to see how availability changes over time.

On the spatial side, the visualization could incorporate aggregation techniques such as clustering or density maps. Rather than displaying only building-level values, the website could highlight spatial patterns of availability across campus, revealing areas with consistently high or low availability. This would support a more global understanding of the data distribution.

Another improvement is to add summary visualizations. For example, histograms or heatmaps could display the distribution of room availability across time (e.g., busiest hours of the day) or across buildings. These views would help users spot patterns, like peak usage times or underused buildings.

Finally, the website could incorporate coordinated multiple views, where interactions in one view (e.g., selecting a time window or building) dynamically update all other visual components. This would strengthen the link between spatial and temporal exploration and provide a more coherent visualization.

Overall, these changes would make the website more useful for exploring and understanding room availability.